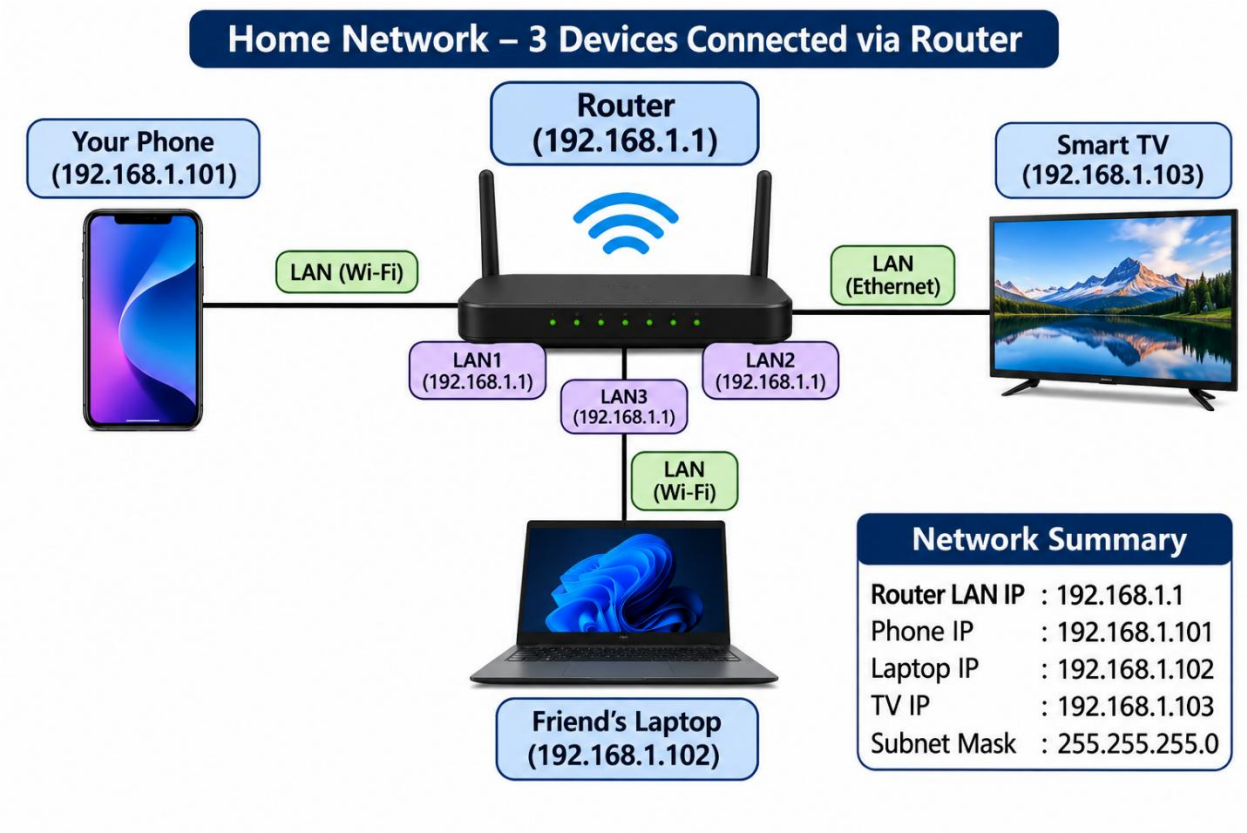
IP routing and routing protocols

Tasks 1 : Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.



Explanation:

All three devices are connected to the same router and belong to the same network, 192.168.1.0/24. The router's LAN interface uses 192.168.1.1 and acts as the default gateway. Devices use their unique IP addresses to communicate over the network.

Conclusion: The diagram shows how a router connects multiple devices within a local area network (LAN).

Task 2 : Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made.

Network Details

- Laptop A: 192.168.1.10/24
- Router R1 LAN: 192.168.1.1/24
- Router R1 WAN: 10.0.0.1/30
- Router R2 WAN: 10.0.0.2/30
- Router R2 LAN: 192.168.2.1/24
- Friend's Computer: 192.168.2.10/24

Packet Journey

1. Laptop A determines that the destination IP address, 192.168.2.10, belongs to a different subnet.
2. It sends the packet to its default gateway, 192.168.1.1.
3. Router R1 checks its routing table and forwards the packet to the next hop, 10.0.0.2.
4. Router R2 checks its routing table and finds that 192.168.2.0/24 is directly connected.
5. Router R2 forwards the packet to the friend's computer.

Routing Decision: Routers select a matching route based on the destination IP address. When multiple routes match, the most specific route (longest prefix match) is preferred.

Task 3 : Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.

Objective: To create a routing table for a fictional food delivery company with branches in three different cities.

The company has three branches: Ahmedabad, Surat, and Mumbai. Each branch uses a different IP subnet. A central router connects the branches through separate WAN links.

Branch	Network Address	Subnet Mask	Gateway
Ahmedabad	192.168.10.0/24	255.255.255.0	192.168.10.1
Surat	192.168.20.0/24	255.255.255.0	192.168.20.1
Mumbai	192.168.30.0/24	255.255.255.0	192.168.30.1

Central Router R0 Routing Table

Destination Network	Subnet Mask	Next Hop	Interface
192.168.10.0	255.255.255.0	10.0.0.2	Serial0/0/0
192.168.20.0	255.255.255.0	10.0.0.6	Serial0/0/1
192.168.30.0	255.255.255.0	10.0.0.10	Serial0/1/0

Explanation: The central router checks the destination IP address against its routing table. It selects the matching network route and forwards the packet to the next-hop router through the specified interface.

Task 4 : Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).

1. Static Routing

Static routing is a method in which a network administrator manually adds routes to a router's routing table.

Advantages

1. **Simple for small networks:** It works well when only a few routes are required.
2. **Low overhead:** It does not need routing protocols to exchange route updates.
3. **More control:** The administrator decides which path traffic should follow.

Disadvantages

1. **Manual configuration:** Every route must be configured by the administrator.
2. **Limited scalability:** Managing many routes becomes difficult in a large network.
3. **No automatic recovery:** If a configured path fails, another route must be configured unless a backup route already exists.

Example: A small food delivery company with two branches can use static routes when its network paths rarely change.

2. Dynamic Routing

Dynamic routing uses protocols such as RIP and OSPF to let routers learn routes and update their routing tables automatically.

Advantages

1. **Automatic route updates:** Routers can learn new routes when the network changes.
2. **Suitable for large networks:** It reduces the need to configure every route manually.
3. **Can adapt to failures:** Routers can select an alternative route when a valid alternative is available.

Disadvantages

1. **More configuration:** Routing protocols require proper setup and troubleshooting.
2. **Uses network resources:** Routers exchange routing information and perform route calculations.
3. **More complex:** Protocol behavior and routing problems can be harder to diagnose.

Example: A food delivery company operating branches in Ahmedabad, Surat, and Mumbai could use OSPF so routers can exchange route information and adapt when a network link fails.

Comparison

Feature	**Static Routing**	**Dynamic Routing**
Configuration	Manual	Protocol-based and automatic route learning
Best suited for	Small, stable networks	Medium and large changing networks
Route updates	Usually manual	Automatic
Example	Fixed route between two branches	OSPF between multiple branches

Task 5 : Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does.

Task 5: Configure OSPF Routing Between Two Routers

Objective

To configure OSPF (Open Shortest Path First) between two routers and understand how each configuration command works.

1. Network Topology

Two routers, R1 and R2, are connected through a WAN link. Each router connects to a separate LAN.

IP Addressing Table

Device	Interface	IP Address	Subnet Mask
PC1	Ethernet	192.168.1.10	255.255.255.0
R1	G0/0	192.168.1.1	255.255.255.0
R1	G0/1	10.0.0.1	255.255.255.252
R2	G0/1	10.0.0.2	255.255.255.252
R2	G0/0	192.168.2.1	255.255.255.0
PC2	Ethernet	192.168.2.10	255.255.255.0

Default Gateway:

- PC1: 192.168.1.1
- PC2: 192.168.2.1

2. OSPF Configuration on Router R1

enable

configure terminal

interface gigabitEthernet 0/0

ip address 192.168.1.1 255.255.255.0

no shutdown

```
exit

interface gigabitEthernet 0/1
 ip address 10.0.0.1 255.255.255.252
 no shutdown
exit
router ospf 1
 network 192.168.1.0 0.0.0.255 area 0
 network 10.0.0.0 0.0.0.3 area 0
exit

end
write memory
```

3. OSPF Configuration on Router R2

```
enable
configure terminal
interface gigabitEthernet 0/0
 ip address 192.168.2.1 255.255.255.0
 no shutdown
exit

interface gigabitEthernet 0/1
 ip address 10.0.0.2 255.255.255.252
 no shutdown
exit
```

```
router ospf 1
 network 192.168.2.0 0.0.0.255 area 0
 network 10.0.0.0 0.0.0.3 area 0
exit

end
write memory
```

Note: Interface names may vary depending on the router model. Use the actual connected interface names in your topology.

4. Explanation of Configuration Commands

Command	Explanation
enable	Enters privileged EXEC mode.
configure terminal	Enters global configuration mode.
interface gigabitEthernet 0/0	Selects the specified router interface.
ip address	Assigns an IPv4 address and subnet mask to the interface.
no shutdown	Enables the interface.
router ospf 1	Starts OSPF routing process 1.
network	Selects interfaces whose IP addresses match the specified network and enables OSPF on them.
0.0.0.255	Wildcard mask matching addresses in the 192.168.1.0/24 network.
0.0.0.3	Wildcard mask matching addresses in the 10.0.0.0/30 network.
area 0	Places the matching interfaces in OSPF backbone area 0.
exit	Leaves the current configuration mode.
end	Returns to privileged EXEC mode.

write memory	Saves the running configuration.

5. How OSPF Works

1. R1 and R2 exchange OSPF Hello packets through their WAN link.
2. If their OSPF settings are compatible and the link is working, they establish an OSPF neighbor relationship.
3. The routers exchange link-state information about their networks.
4. Each router calculates the best available route using OSPF's cost metric.
5. R1 learns how to reach 192.168.2.0/24, while R2 learns how to reach 192.168.1.0/24.

This allows PC1 and PC2 to communicate across different subnets without manually configuring a static route for each remote LAN.

6. Verification Commands

Run the following commands on both routers:

show ip ospf neighbor

show ip route ospf

show ip protocols

show ip interface brief

Expected Results:

- show ip ospf neighbor displays the OSPF neighbor if adjacency is established.
- show ip route ospf displays routes learned through OSPF. R1 should learn 192.168.2.0/24, and R2 should learn 192.168.1.0/24.
- show ip protocols displays routing protocol information.
- show ip interface brief shows the interface status and IP addresses.

To test connectivity, open PC1's command prompt and run:

ping 192.168.2.10

If addressing, interfaces, OSPF adjacency, and PC settings are correct, PC1 should be able to reach PC2.